# RM9: Accepted lane-flow kernel audit and the next physical-resource decision

Method-first ramp-metering review

2 October 2026

**Disposition.** The user accepted the previous handout and RM9-UG2.A for analytic review. RM9-X1–X3 remain deferred. This audit establishes conditional mathematical consequences of the accepted kernel, not traffic calibration, an identical-history action certificate, or independently useful gradients. A new gate, RM9-UG3, reopens only the resource interpretation of the residual longitudinal budgets in RM9-G2. No equation has been changed silently.

## 1 Scope, notation and what acceptance established

Use two physical lanes per local longitudinal cell. Mass $n_{i\ell}$ is continuous vehicle equivalents; $L$ is cell length, $h$ the timestep, and $N_{i\ell}^{\max} = Lk_{i\ell}^{\mathrm{jam}}$ its finite storage. Distinguish total observed cell mass $N_i = n_{i1} + n_{i2}$ from lane maximum storage. Let $c_\ell = v_\ell/L$, $\gamma_\ell = w_\ell/L$. Ordinary lane sending/receiving are

$$D_\ell(n) = \min(c_\ell n, C_\ell), \qquad S_\ell(n) = \min(C_\ell, \gamma_\ell(N_\ell^{\max} - n)). \tag{1}$$

The accepted step bounds $hc_\ell \le 1$, $h\gamma_\ell \le 1$ imply $hD_\ell(n) \le n$ and $hS_\ell(n) \le N_\ell^{\max} - n$. The bounded discretionary propensity generates at most one positive direction in a two-lane cell, with actual lateral rate $\lambda_{\ell j} \le \min(D_\ell(n_\ell), S_j(n_j))$. After conservative exchange,

$$n_\ell^L = n_\ell + h(\lambda_{j\ell} - \lambda_{\ell j}). \tag{2}$$

The accepted kernel then limits longitudinal movement by

$$\bar{D}_\ell = \min\{D_\ell(n_\ell^L), D_\ell(n_\ell) - \lambda_{\ell j}\}, \tag{3}$$

$$\bar{S}_\ell = \min\{S_\ell(n_\ell^L), S_\ell(n_\ell) - \lambda_{j\ell}\}. \tag{4}$$

Ordinary forward interfaces and the Default priority merge use these budgets. An upstream meter still bounds upstream release; it is not automatically a bound on instantaneous downstream entry. Ramp and external waiting need their corresponding conserved states. Exact local geometry and observer support are not yet frozen by handout acceptance.

**Preserved decisions.** Common bounded open-loop action, complete all-stream time through clearance, strong aggregate density/flow/speed/count plus ramp history, known parameters/ forcing, and Default positive structural priorities remain unchanged. No acceleration, end-position rule, extra actuator or source repair is opened.

## 2 What passes mathematically; a lossless information coordinate

The lateral step keeps each lane in $[0, N_\ell^{\max}]$ and conserves their sum. Residual budgets are nonnegative. With accepted forward/merge incoming rate $f_{i\ell}^{\text{in}} \le \bar{S}_{i\ell}$ and outgoing rate $q_{i\ell} \le \bar{D}_{i\ell}$,

$$n_{i\ell}^{+} = n_{i\ell}^{L} + h(f_{i\ell}^{\text{in}} - q_{i\ell}) \quad \Longrightarrow \quad 0 \le n_{i\ell}^{+} \le N_{i\ell}^{\max}. \tag{5}$$

Every internal accepted transfer appears once with each sign. Thus the operator is conservative and storage-bounded, conditional on corresponding ramp/boundary balances. This is a proof of an update, not complete-task geometry or physical-fidelity acceptance.

For two lanes, one lane coordinate together with road total retains the entire cell's lane information:

$$z_i = (N_i, b_i), \qquad b_i = n_{i1}, \qquad (n_{i1}, n_{i2}) = (b_i, N_i - b_i). \tag{6}$$

Equivalently use $\delta_i = n_{i1} - n_{i2}$, giving $n_{i1} = (N_i + \delta_i)/2$, $n_{i2} = (N_i - \delta_i)/2$. This is a lossless reparameterization of the accepted two-lane reference, not compression of several longitudinal cells into one zone-wide scalar, proof that contrast is hidden, or necessity of individual vehicle identity.

With positive density denominators and fixed strict branches of min/max, the propensity, conservation and priority operators are differentiable compositions. At a switch or empty-state denominator, an ordinary derivative is not asserted. Correct branch algebra alone does not verify the full task's gradient or independent one-step usefulness.

## 3 The new consequence: an implicit forward-service debit

Consider an ordinary cell away from the ramp-entry interface. Both pre/post sending functions are on uncapped free-flow branches, and downstream residual receiving is sufficient for both forward offers. Suppose lateral transfer $\lambda > 0$ goes from lane $d$ to lane $t$:

$$n_d^L = n_d - h\lambda, \qquad n_t^L = n_t + h\lambda. \tag{7}$$

The accepted residual law yields

$$\bar{D}_d = \min(c_d n_d - h c_d \lambda, c_d n_d - \lambda) = c_d n_d - \lambda, \tag{8}$$

$$\bar{D}_t = \min(c_t n_t + h c_t \lambda, c_t n_t) = c_t n_t, \tag{9}$$

since $h c_d \le 1$. Consequently the total longitudinal rate is

$$Q = c_d n_d + c_t n_t - \lambda. \tag{10}$$

The comparison with lateral movement disabled has the same initial state and ordinary sending parameters: $Q_0 = c_d n_d + c_t n_t$. This is a local ablation diagnostic, not two identical-observation histories.

# 4 Meaning, a hand calculation, and source limits

Equation (10) does not require receiving congestion, lane-speed equality, a speed-dependent ramp cost or an explicit capacity-drop coefficient. The donor loses one unit of forward-rate budget per unit lateral-rate transfer; the target cannot increase sending beyond its original budget. The debit persists as $h \to 0$ at fixed cell state/length/propensity. It is therefore not merely a disappearing finite-timestep truncation effect.

**Hand-calculated feasible branch.** Use illustrative consistent units, $L = 1$, $h = 1/4$, $c_d = 1$, $c_t = 2$, $n_d = 6$, $n_t = 2$, $\mu = 1/2$, $\kappa = 1$. Select finite capacities, storage and downstream receiving sufficiently large that all stated branches are strict. The propensity and transfer are

$$A = \frac{1}{2}\frac{6-2}{6+2} = \frac{1}{4}, \qquad \lambda = Ac_d n_d = \frac{3}{2}. \tag{11}$$

Post-lateral masses are 45/8 and 19/8. Their ordinary sending rates are 45/8 and 19/4, but the residual budgets impose

$$q_d = \min(45/8, 6 - 3/2) = 9/2, \qquad q_t = \min(19/4, 4) = 4, \qquad Q = 17/2. \tag{12}$$

The no-lateral initial offer is $Q_0 = 10$; the difference is exactly 3/2. Mass is still conserved. No traffic trajectory or numerical experiment was run. The lane mobilities are different, avoiding a symmetry selected to conceal state.

**Scientific interpretation.** A common movement resource can be a deliberately stipulated model: sideways and forward transfers compete for that resource. This is not mathematically impossible. But the accepted proposal's conservation proof does not establish that within-cell lane changing consumes longitudinal service one for one in physical traffic. The earlier phrase "no capacity-drop penalty" must be read narrowly as no *explicit* acceleration/capacity-drop law. It is not a no-throughput-cost claim: (10) demonstrates an implicit debit. The earlier explanation was insufficiently explicit about its magnitude.

Source FB0030 [1], printed equations 20–22, adjusts sending with both lateral inflow and outflow; the accepted target's old-sending cap is not that complete source coupling. Its unresolved branches are not repaired here. RM0018 [2] specifies sharing at an input-to-output node; it does not establish the debit for a transverse within-cell transfer. Neither inspected source validates (10) as this task's physical lane-changing cost.

A meter perturbation could alter lane contrast, hence $\lambda$ and this debit. A resulting internal gradient may be correct for the stipulated resource law, but does not independently prove a real metering benefit, hidden information, or total-time improvement through common clearance. Different final departure history is still required; a local flow difference is not sufficient.

# 5 Conditional strong-observer reconstruction, not a certificate

This diagnostic retains an *independent* speed channel rather than silently replacing speed with total flow divided by density. On the ordinary strict free-flow branch above, suppose known distinct lane speeds $v_1 \neq v_2$ are also the lane passage speeds at the co-supported observation. Let the observer receive $N = n_1 + n_2$, $Q = q_1 + q_2 > 0$ and passage-weighted speed

$$V = \frac{v_1 q_1 + v_2 q_2}{Q}. \tag{13}$$

Then the two individual realized flows reconstruct algebraically:

$$q_1 = \frac{Q(V - v_2)}{v_1 - v_2}, \qquad q_2 = \frac{Q(v_1 - V)}{v_1 - v_2}. \tag{14}$$

Define $x_\ell = q_\ell / c_\ell$ and $\epsilon = N - x_1 - x_2$. Assume both lane masses are positive, and only the discretionary coefficient $\mu \in [0, 1]$, $\kappa \in (0, 1]$ operates. If $\lambda = 0$, then $\epsilon = 0$ and $n_\ell = x_\ell$. If lateral flow is active, (9) gives

$$\epsilon = \lambda / c_d > 0, \qquad x_d = n_d - \epsilon, \qquad x_t = n_t. \tag{15}$$

The active propensity is bounded by the unbiased density difference:

$$\epsilon = A n_d \leq \frac{n_d(n_d - n_t)}{n_d + n_t} < n_d - n_t, \qquad n_d > n_t > 0. \tag{16}$$

Hence $x_d > x_t$ and the donor is identifiable from normalized realized flows. Recover the original state without knowing the hidden branch in advance:

$$d = \operatorname*{argmax}_{\ell \in \{1,2\}} x_\ell, \qquad n_d = x_d + \epsilon, \quad n_t = x_t. \tag{17}$$

At zero transfer, $n = x$ directly; no argmax is needed. Thus, under this explicitly stated speed map and branch, even a snapshot of aggregate mass/flow/independent speed identifies lane occupancy. A full equal history cannot conceal different current lane masses in this subcase.

This is a **conditional reviewer theorem**: the accepted kernel does not itself supply a calibrated independent detector-speed map, so the diagnostic is not adopted as the complete observer. It excludes capacity/supply-bound cells, empty recipients, equal lane speeds, different observation supports and changing lane passage speeds. No conclusion of global congested-state or field reconstruction follows. No weaker speed channel or exact history pair was substituted to make the research answer positive.

## 6 The concrete alternative for RM9-UG3.B

**Proposed, not adopted.** Keep the bounded lateral exchange and its finite mass/storage proof. Treat it as a transverse occupancy transfer, separate from the longitudinal face's flow capacity. After it, recompute

$$D^B_{i\ell} = D_{i\ell}(n^L_{i\ell}), \qquad S^B_{i\ell} = S_{i\ell}(n^L_{i\ell}). \tag{18}$$

Ordinary longitudinal flows use these limits. Mainline and ramp offers still share the same longitudinal receiving budget via the Default allocator. No lateral transfer is also charged against the original longitudinal-rate budget. This changes the physical resource interpretation; it is not a clerical removal of a redundant min or an approved correction.

With incoming longitudinal/ramp rate $f^B \le S^B(n^L)$ and outgoing $q^B \le D^B(n^L)$, the same step bounds give

$$n^+ = n^L + h(f^B - q^B) \in [0, N^{\max}]. \tag{19}$$

Internal transfer signs still cancel. Ramp/external balances remain mandatory. Total accepted transverse-plus-longitudinal incoming rate need not be below the *original* longitudinal $S$; it is not being claimed to use one flow-capacity resource. Both stages do respect mass/storage at their respective states. This distinction resolves the earlier "double capacity" concern only by explicitly changing which resource the two movements share. It requires approval.

On the same uncapped, unimpeded ordinary branch,

$$Q_B = c_d(n_d - h\lambda) + c_t(n_t + h\lambda) = Q_0 + h(c_t - c_d)\lambda. \tag{20}$$

There is no universal $-\lambda$ forward debit; changes follow the redistributed state and differing lane mobilities. At fixed state as $h \to 0$, this local step rate approaches $Q_0$. This is not proof of perfect traffic fidelity, a globally best unmetered policy, or a positive hidden-state result. The propensity's physical spatial scale, step splitting, full observer and geometry still require review. No restored source branch, acceleration state or deferred X1–X3 mechanism is added. Only the residual flow-capacity debits are proposed for reconsideration.

## 7 Gate and claim status

RM9-UG2 is closed by the user's acceptance, preserved in its approval record. RM9-UG3 explicitly reopens the smallest RM9-G2 resource contract: retain the accepted common movement resource as a declared debit model (A), or adopt separate transverse/longitudinal resources with conservative recomputation for the continued review (B, provisional recommendation). Neither choice proves hidden-information necessity. The full observation and initialization contract is unfinished; no feasible equal-history useful-action pair, whole-chain gradient check or independent one-step usefulness is established. No experiment, implementation or successor is open.

## Primary source boundary

1. Existing FB0030, final primary PDF, printed pp. 1762–1766 and 1768–1769, source PDF. Selected transfer/coupling and sensor-support sections rechecked. No full source scheme repaired or validated.
2. Existing RM0018, Wright et al. (2017), published primary author-hosted node paper, Section 3.1, DOI 10.1016/j.trb.2017.09.001. It establishes node-allocation antecedents, not within-cell physical debit.

Conservation, free-flow debit, hand calculation, conditional reconstruction and alternative-operator proofs are reviewer derivations under explicit conditions. They are not joint source findings or a combined literature proof of the core review questions.